

A SQL-Based Analysis of Sales and Business Performance at Gayanara Fashion Store



[Zata Maitsaa Syahada] • SQL Project Documentation

From raw data to evidence-based insights

Project at a Glance

A one-slide summary for someone who has 30 seconds to understand the project.

THE CONTEXT

Gayanara Fashion Store generates transaction data that can be analyzed to understand sales performance, customer behavior, product performance, and operational activities.

THE OBJECTIVE

To analyze sales and business performance using SQL and identify actionable insights across customers, products, promotions, logistics, geography, ratings, and payment behavior.

THE OUTPUT

A SQL-based analysis that identifies key business patterns, customer segments, product performance, promotional profitability, operational issues, and revenue opportunities.

KEY QUESTIONS

- 01 Customer -- Who are the most valuable customers based on RFM?
- 02 Product & Inventory -- Which products perform best, and which have idle inventory?
- 03 Operations & Profitability -- Are promotions profitable, and which courier has the most returns?
- 04 Market & Customer Experience -- Where is revenue highest, what ratings are common/anomalous, and which payment methods are preferred?

Dataset Overview

Keep this factual and compact.

[5]

TABLES

[10.586]

TOTAL RECORDS

[44]

VARIABLES

[Relational]

DATA STRUCTURES

SOURCE

Gayanara Fashion Store relational dataset
<https://ngulikdata.com/datasets/gayanara>

KEY VARIABLES

customers — customer_id, name, city
products — product_id, name, category, price_idr
orders — order_id, customer_id, total_amount_idr, order_status
order_items — order_id, product_id, subtotal_idr
reviews — order_id, product_id, customer_id

Data Quality & Cleaning

Show the problems first, then the cleaning approach.

DUPLICATES

No duplicate records found

Checked across the five related tables using key-based duplicate checks.

Action: No removal required.

DATA TYPE

Date stored as text

Date fields were initially stored as text values.

Action: Converted text values into the appropriate DATE format for analysis.

CATEGORY INCONSISTENCIES

inconsistent product categories

Product category values contained inconsistent naming or formatting.

Action: Standardized category values into consistent categories.

CLEANING PRINCIPLE

Ensure data consistency and analysis readiness while preserving the meaning of the original data.

SQL Cleaning Technique

```
update prod_dc
set category =
case
when category IN('shirt','T-shirt','kaos') then 'Kaos'
when category IN('Accessories', 'aksesoris') then
'Aksesoris'
when category IN('Pants', 'celana') then 'Celana'
when category IN('Jacket', 'jaket') then 'Jaket'
else category
end;
```

TECHNIQUE

CASE WHEN + UPDATE

PURPOSE

Standardize inconsistent product category names into consistent categories.

VALIDATION

Compared category values before and after the transformation to ensure consistent naming.

Category Standardization

Standardizing inconsistent product categories for reliable analysis.

	category
▶	Accessories
	Aksesoris
	Celana
	dress
	Jacket
	Jaket
	Kaos
	kemeja

Inconsistent category naming and capitalization

	category
▶	Accessories
	Aksesoris
	Celana
	dress
	Jacket
	Jaket
	Kaos
	kemeja

Standardized category values for consistent analysis

Result: Product categories were standardized into consistent labels, enabling reliable grouping and aggregation in subsequent analysis.

Exploratory Data Analysis

1. Revenue by City

```
select c.city, sum(o.total_amount_idr)
total_revenue, count(o.order_id) 'total
penjualan'
from cust_dc c
join ord_dc o
on c.customer_id = o.customer_id

where order_status NOT IN ('cancelled',
'returned')

group by c.city
order by sum(o.total_amount_idr) desc
limit 10;
```

ANALYSIS APPROACH

Revenue is calculated from non-cancelled and non-returned orders, grouped by city and ranked by total revenue.

city	total_revenue	total penjualan
Medan	84676287	170
Depok	78838753	158
Semarang	74719258	145
Bekasi	73193453	156
Balikpapan	71122758	133

KEY FINDING

Medan generated the highest total revenue among the analyzed cities, followed by Depok and Semarang.

Exploratory Data Analysis

2. Revenue by Province

```
select c.province,  
sum(o.total_amount_idr) total_revenue,  
count(o.order_id) 'total penjualan'  
from cust_dc c  
join ord_dc o  
on c.customer_id = o.customer_id  
  
where order_status NOT IN ('cancelled',  
'returned')  
  
group by c.province  
order by sum(o.total_amount_idr) desc  
limit 10;
```

ANALYSIS APPROACH

Revenue is calculated from non-cancelled and non-returned orders, grouped by city and ranked by total revenue.

province	total_revenue	total penjualan
Jawa Barat	258059297	513
Jawa Timur	135570927	275
Sumatera Utara	84676287	170
Jawa Tengah	74719258	145
Kalimantan Timur	71122758	133

KEY FINDING

West Java generated the highest revenue at IDR 258.06 million, followed by East Java at IDR 135.71 million.

Exploratory Data Analysis

3. Best Selling Products

```
with penjualan_valid as
(
  select oi.product_id, oi.order_id, o.order_status, oi.subtotal_idr
  from ord_dc o
  join oi_dc oi
    on o.order_id = oi.order_id
  where o.order_status NOT IN ('cancelled', 'returned')
)
select p.`name`, p.stock, sum(pv.subtotal_idr) total_revenue from penjualan_valid pv
join prod_dc p
  on pv.product_id = p.product_id
group by p.`name`, p.stock
order by sum(pv.subtotal_idr) desc
limit 5;
```

ANALYSIS APPROACH:

CTE + JOIN + Aggregation

A CTE was used to isolate valid sales by excluding cancelled and returned orders. The result was then joined with product data and aggregated by product to calculate total revenue.

name	stock	total_revenue
Dress Midi Floral NusaBrand	57	14975000
Shirt Slim Fit Tropika Style	26	14970000
Celana Jogger Tropika Style	86	13777000
Kemeja Oxford NusaBrand	32	13725000
Dress Bodycon Pesona Indo	0	13178000

KEY FINDING

Dress Midi Floral NusaBrand generated the highest revenue at IDR 14.98 million, closely followed by Shirt Slim Fit Tropika Style at IDR 14.97 million.

Exploratory Data Analysis

4. Potential slow-moving inventory

```
select p.`name`, p.stock,  
count(oi.product_id) total_terjual  
from prod_dc p  
join oi_dc oi  
  on oi.product_id = p.product_id  
group by `name`, p.stock, p.product_id  
order by total_terjual asc, p.stock desc;
```

ANALYSIS APPROACH:

Sales were aggregated by product and compared with the available stock to identify products with low sales volume relative to inventory.

name	stock	total_terjual
Celana Jogger Ratu Mode	239	7
Celana Kulot Senja Wear	39	7
Jaket Varsity SandangIndo	407	8
Jaket Varsity Pesona Indo	225	9
Kaos Oversize Ratu Mode	141	9

KEY FINDING

Jaket Varsity SandangIndo had 407 units in stock but only 8 units sold, making it a notable slow-moving product.

Exploratory Data Analysis

5. Promotion Effectiveness Analysis

```
select
  case
    when o.discount_amount_idr > 0 then 'pakai promo'
    else 'tanpa promo'
  end as keterangan_promo,
  count(o.order_id) total_pembelian,
  sum(o.total_amount_idr) total_revenue,
  avg(o.total_amount_idr) as rerata_belanja
from ord_dc o
where o.order_status NOT IN ('cancelled', 'returned')
group by keterangan_promo
order by total_revenue desc;
```

ANALYSIS APPROACH:

Categorized active orders based on promotion usage using CASE WHEN, then aggregated total orders, total revenue, and average spending per category.

keterangan_promo	total_pembelian	total_revenue	rerata_belanja
tanpa promo	1496	768689000	513829.5455
pakai promo	1025	496651379	484537.9307

KEY FINDING

Transactions *without promo* generated higher total revenue (IDR 76.8M) and average spending (IDR 51.3K) compared to transactions *with promo* (IDR 49.6M and IDR 48.4K).

Exploratory Data Analysis

6. RFM Customer Segmentation

```
with rfm as
(
select customer_id,
datediff('2025-03-01', MAX(order_date)) as recency,
count(order_id) as frequency,
sum(total_amount_idr) as monetary
from ord_dc
where order_status NOT IN ('cancelled', 'returned')
group by customer_id
)
select *,
case
when recency < 300 and frequency >= 2 and monetary > 1000000 then 'VIP'
when recency >= 300 and frequency >=1 and monetary between 500000 and 1000000 then 'Pelanggan casual'
when recency > 1000 and frequency = 1 then 'Perlu retargeting'
end as Klasifikasi_Pelanggan
from rfm;
```

ANALYSIS APPROACH:

Built a CTE (Common Table Expression) named rfm to calculate Recency, Frequency, and Monetary metrics for active customers, then applied a CASE WHEN statement to segment them into specific customer classifications.

customer_id	recency	frequency	monetary	Klasifikasi_Pelanggan
CUST0474	130	6	7803360	VIP
CUST0478	3	4	2231170	VIP
CUST0170	28	6	2939760	VIP
CUST0200	296	2	1521669	VIP
CUST0225	49	6	1544654	VIP

KEY FINDING

Successfully segmented customers based on their purchasing behavior (Recency, Frequency, Monetary), identifying high-value segments like VIP customers (e.g., CUST0474 with 6 transactions and IDR 7.8M monetary value) for targeted retention strategies.

Exploratory Data Analysis

7. Courier and Returned Orders

```
select courier,  
count(order_id) as 'total barang yg dikembalikan',  
order_status  
from ord_dc  
where order_status = 'returned'  
group by courier  
order by count(order_id) desc;
```

ANALYSIS APPROACH:

Filtered orders with a returned status, then grouped and counted the total returned items by each courier service, ordered from highest to lowest..

courier	total barang yg dikembalikan	order_status
J&T	54	returned
JNE	43	returned
SiCepat	35	returned
Pos Indonesia	11	returned
Anteraja	8	returned

KEY FINDING

J&T recorded the highest number of returned items (54), followed by JNE (43) and SiCepat (35), while Anteraja had the lowest returns (8)

Exploratory Data Analysis

9. Q4 Sales Performance by Year

```
with month_cte as
(
select MONTH(order_date) months,
sum(total_amount_idr) total_2024_q4,
avg(total_amount_idr) rerata_pembelian
from ord_dc
where order_date NOT IN ('cancelled', 'returned')
group by order_date
having order_date between '2024-10-01' and '2024-12-31'
order by order_date asc
), last_3 as
(
select months, sum(total_2024_q4) as sum_each_month from month_cte
group by months
)
select months,
sum(sum_each_month) as 'total revenue 2024' from last_3
group by months;
```

ANALYSIS APPROACH:

Utilized multiple CTEs to filter active orders, extract months from order_date, filter for Q4 months (October–December), and aggregate total revenue grouped by month.

2022

months	total revenue 2022
10	23651346
11	21344757
12	21968852

2023

months	total revenue 2023
10	41292906
11	33265206
12	44456053

2024

months	total revenue 2024
10	48000198
11	58535660
12	47427213

KEY FINDING

Q4 revenue showed a consistent upward growth trend from 2022 to 2024, with peak performance generally occurring in November (e.g., reaching IDR 58.5M in November 2024).

Exploratory Data Analysis

10. Store Rating Distribution Analysis

```
select rating,  
count(*) as TOTAL  
from rev_dc  
group by rating  
order by 2 desc;
```

ANALYSIS APPROACH:

Counted the total number of reviews from the review table (rev_dc) grouped by each rating score (1 to 5), then sorted the results in descending order by the total count.

rating	TOTAL
5	666
4	430
3	234
2	104
1	66

KEY FINDING

The store's ratings are predominantly positive. Ratings of 5 (666 reviews) and 4 (430 reviews) form the vast majority, while low ratings of 1 and 2 are minimal (66 and 104 reviews).

Key Business Insights

The most important slide: move from numbers to meaning.

01 Geographic Concentration

West Java and major cities like Medan and Depok drive the vast majority of revenue.

02 Promotional Strategy Reevaluation

Non-promo transactions currently yield higher average spending and total revenue than promo transactions.

03 High Inventory Capital Lock

Certain items suffer from severe low turnover despite massive stock levels.

04 Strong Customer Loyalty & Satisfaction

The store maintains an overwhelmingly positive reputation and stable seasonal growth.

Business Implications

What can a stakeholder reasonably take from the analysis?

WHAT TO MONITOR

Track Q4 seasonal spikes (especially November surges) and monitor high-value VIP customer segments identified via RFM.

WHAT TO INVESTIGATE

Analyze why discount campaigns underperformed compared to non-promo sales, and evaluate root causes behind high return volumes on couriers like J&T.

WHAT TO CONSIDER

Implement promotional restructuring to target specific margins, optimize inventory holding costs for slow-moving items (e.g., varsity jackets), and launch retention campaigns for VIP segments.

Conclusion

Close the story in three lines.

DATA

Analyzed 10,586 records across 5 relational tables covering sales, product inventory, promotions, customer RFM, logistics, and reviews.

INSIGHT

Revenue is heavily concentrated in specific regions (West Java/Medan), discount strategies currently underperform compared to non-promo sales, and inventory imbalances exist with slow-moving stock.

VALUE

Empowers management to reallocate marketing budgets, refine discount mechanics, manage dead stock, and improve logistical efficiency.

Project Repository

Keep the PDF concise; keep the detailed SQL in GitHub.

01 SQL

<https://github.com/zata-syahada/sql-data-analysis-portfolio/tree/main/2-gayanara-fashion-store>

02 DATA

<https://ngulikdata.com/datasets/gayanara>

03 DOCUMENTATION

<https://github.com/zata-syahada/sql-data-analysis-portfolio/tree/main/2-gayanara-fashion-store>

<https://github.com/zata-syahada/sql-data-analysis-portfolio/tree/main>

END

THANK YOU

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